



Fullstack Web Development

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Web Development Fundamentals

- Understand how websites work and the roles of HTML, CSS, and JavaScript in web development.
- Learn how the internet functions, including how websites are hosted and served.
- Gain hands-on experience with Git, GitHub, and version control.
- Develop key troubleshooting and debugging skills to apply to your projects.

HTML 5

- HTML Expedition: Learn HTML syntax to structure your digital world.
- Base Camp Setup: Use HTML boilerplate and understand doctypes.
- Nesting Mastery: Apply indentation and nesting for clean code.
- Tag Treasures: Use HTML tags for headings and paragraphs.
- Conquer Lists: Create ordered and unordered lists.
- Capture Moments: Insert images with the tag.
- Link Builder: Create page links with anchor tags.
- Multi-Page Adventure: Design interconnected web pages.
- Navigate HTML: Use best practices for effective coding.

CSS

- CSS Basics: Discover how cascading style sheets bring your website to life.
- Selector Skills: Master CSS selectors and properties for styling.
- Inline to External: Learn to apply CSS using inline, internal, and external methods.
- Best Practices: Follow CSS coding best practices for clean styles.
- Specificity Secrets: Understand CSS specificity and inheritance rules.
- Box Model Blueprint: Dive into the CSS Box Model for layout control.
- Inspect & Diagnose: Use developer tools to troubleshoot CSS.
- Positioning & Display: Explore CSS positioning and display techniques. Font Styling: Style text with CSS and web-safe fonts.
- Responsive Magic: Implement responsiveness using media queries.
- Float & Clear: Utilize float and clear properties for layout.
- Selector Mastery: Combine CSS selectors and grasp their priorities.

FLEXBOX

- Flexbox Origins: Learn why Flexbox was introduced to CSS.
- Display Flex: Understand display: flex for flexible layouts.
- Direction Control: Master flex-direction to control layout flow.
- Align & Justify: Position child items using alignment and justification.
- Sizing & Distribution: Use the flex shorthand to control child item sizing.

GRIDS

- Grid vs. Flexbox: Explore differences and strengths of Grid layout.
- Grid Display: Use display: grid for structured designs.
- Grid Sizing: Create and arrange grids with tracks and gaps.
- Positioning: Place items across rows and columns.
- Combine Forces: Mix Grid with Flexbox for complex layouts.

BOOTSTRAP

- Native vs. Frameworks: Explore the differences between CSS tooling and external frameworks.
- Bootstrap Installation: Learn to add Bootstrap to your website for rapid styling.
- 12-Column Layout: Master the Bootstrap grid system for flexible layouts.
- Components Galore: Use Bootstrap's buttons, carousels, cards, and navbars.
- Icon Access: Incorporate designer icons from Bootstrap's collection.
- Quick Builds: Use code snippets, examples, and templates to create stunning, mobile-first websites.

WEB DESIGN

- Design Principles: Master the fundamentals of web design, UI, and UX.
- Color Theory: Explore modern color palettes for a professional website look.
- Typography: Choose stylish, free-for-commercial-use fonts for a user-friendly design.
- UI Focus: Learn to manage user attention using key UI design principles.
- UX Mastery: Design user-friendly websites by understanding core UX concepts.

DOCUMENT OBJECT MODEL(DOM)

- HTML Tree: Explore the tree structure of HTML-based websites.
- DOM Navigation: Traverse the document using object notation.
- Separation of Concerns: Apply coding best practices for clean, maintainable code.
- Element Manipulation: Change HTML elements using your DOM mastery.

JAVASCRIPT ES6

- Language Showdown: Compare JavaScript with scripting, markup, and stylesheet languages.
- JavaScript Overview: Explore JS and its versatile uses in web development.
- Coding Tools: Learn to work with code editors and IDEs.
- Syntax & Data Types: Understand variables (let, const), data types, and operators.
- Control Structures: Master if/else, loops (for, while, for...of, for...in) for code flow.
- Functions: Explore function declarations, scope, closures, arrow functions, and higher-order functions.
- Array Mastery: Create, manipulate, and use methods (map, filter, reduce, slice, concat).
- Spread Syntax: Utilize ... for array and function call expansion.
- OOP in JS: Create objects using literals, constructors, and classes; understand this and super.
- Object Handling: Use Object.keys, Object.values, and Object.assign for manipulation.

REACT.JS

- React Adventure: Dive into front-end development with React.
- Component Mastery: Learn when and how to use React components.
- Prop Passing: Discover how to pass and use props effectively.
- JSX Magic: Write and understand JSX syntax for dynamic UI.
- React DOM: Explore the React DOM for rendering components.
- State Management: Master state handling in React.
- Hook Exploration: Learn about React Hooks for enhanced functionality.
- Conditional Rendering: Implement conditional logic in your components.
- Component Types: Understand the difference between class and functional components.

GIT, GITHUB, AND VERSION CONTROL

- VersionControl Basics: Use Git for version control and seamless collaboration.
- Fork, Branch & Clone: Master forking, branching, and cloning for efficient workflows.
- Remote Repos: Utilize GitHub as your remote code repository.
- Change Management: Learn to checkout and roll back changes using Git.
- Git with Xcode: Integrate Git and GitHub into your Xcode projects.

NODE.JS

- Back-End Basics: Explore the components of back-end development using an MVC framework.
- OOP in Node: Apply data types, objects, methods, and classes to back-end development.
- Server-Side JavaScript: Dive into JavaScript for server-side programming.
- Node Command Line: Use Node.js commands for back-end tasks.
- NPM Power: Manage packages with Node Package Manager (NPM).
- Build Processes: Learn JavaScript build processes for efficient development.
- Event Loop: Understand Node's event loop and emitters for asynchronous operations.
- File System: Interact with the file system using Node.js.
- Modular Magic: Work with modules for organized, reusable code.
- Native Drivers: Utilize native Node.js drivers for various database interactions.

EXPRESS.JS

- Express Setup: Learn how to install and use Express in Node applications.
- Server Creation: Build servers using Node and Express.
- RESTful Routing: Master routing to handle requests with Express.
- Middleware Magic: Implement middleware for enhanced Node applications.

APPLICATION PROGRAMMING INTERFACE (API)

- API Basics: Explore what APIs are and how they function.
- Deep Dive into HTTP: Understand the role of HTTP in API communication.
- API Calls: Learn how to make API requests effectively.
- Read the Docs: Navigate and understand API documentation.
- API Authentication: Grasp basic authentication methods for secure access.
- Server Communication: Implement server-to-server interactions. Data Formats: Compare JSON and XML for data exchange.

MONGO DB

- Introduction to MongoDB: Discover the world of NoSQL databases with MongoDB.
- Setting Up: Learn how to install and set up MongoDB for your projects.
- Database Structure: Explore collections and documents as the building blocks of MongoDB.
- CRUD Operations: Master Create, Read, Update, and Delete operations in MongoDB.
- Querying the Database: Use queries to search and filter data effectively.
- Schema Design: Understand schema-less design and data modeling for flexibility.
- Aggregation: Harness aggregation pipelines to process and analyze data.
- Indexes: Learn to create indexes for faster search and query performance.
- Relationships: Implement relationships between data using references and embedding.
- Integration with Node.js: Connect your MongoDB database to Node.js applications.

PROJECTS

- Personal Portfolio Website
- QR Code Generator
- Building a React Advice App with API Integration
- Weather App with React
- BMI Calculator App with React
- Currency Converter App with React
- Digital Clock with React
- Strong Password Generator in React
- Quiz App with React
- CRUD Application with React.js Frontend and Node.js Backend

CHALLENGES AND INTERACTIVE LEARNING

- Each topic features engaging challenges to provide hands-on, real-world learning experiences.
- These activities start easy, building up in difficulty to reinforce concepts.
- Students receive instant feedback, collaborate in groups, and participate in friendly competitions to boost skills and confidence.
- Advanced learners can "challenge a mentor" for deeper insights, ensuring a dynamic, enjoyable, and practical learning journey.
- This approach makes the course interactive, motivating, and memorable.

AI TOOLS FOR MODERN DEVELOPERS

- In addition to MERN stack development, students will learn to use powerful AI tools that improve productivity, design, and coding efficiency.

AI TOOLS COVERED

- Claude - AI assistant for coding help, debugging, and documentation writing
- Vercel AI-powered deployment platform for hosting and scaling full-stack applications.
- Antigravity-AI development tool for faster coding workflows and automation.
- Figma-AI-assisted UI/UX design tool to create modern website and app designs.
- Miro AI collaboration and brainstorming tool for planning projects, system design, and workflows.

WHAT STUDENTS WILL LEARN

- Using AI for code generation and debugging
- Designing UI prototypes before development
- Deploying MERN applications using modern cloud tools
- Project planning and collaboration with AI tools
- Increasing developer productivity using AI workflows

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